(Please write your Enrollment Number) Enrollment No. 04801172022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Off Line)
Course Name: B. Tech, Semester: 7
(December, 2025)

Subject Code: BAI 417	Subject: Multimodal Data Analysis
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	(a) Explain the various types of Language Models.		(CO1)
	(b) Explain the Subword Tokenization.		(CO1)
	(c) Explain the concept of Spectrogram in Short-Term Analysis?		(CO2)
	(d) Define the need for Windowing in Short-Term Analysis.		(CO2)
	(e) Explain the concept of Edge Detection in Digital Image and Video Processing.		(CO3)
	(f) Describe point processing & neighborhood processing in Video processing.		(CO3)
	(g) Define Co-learning and its application in Multimodal Data Integration.		(CO4)
	(h) Give some applications of Multimodal Recommendation Systems.		(CO4)
	UNIT-I		CO Mapping
Q2	(a) In a large dataset of text, it is observed that k=10, β=0.5. Estimate the vocabulary size if the total number of words in the text is 1,000,000.	(2.5)	(CO1)
	(b) Write the examples and applications of Zipf's Law & Heap's Law.	(5)	
	(c) A collection of text has a total of 300,000 words. The constants k= 15 and , β=0.6. If the total word count increases to 600,000, estimate the new vocabulary size.	(2.5)	
Q3	(a) What are the different Multimodal Learning Techniques.	(2.5)	(CO1)
	(b) What are different combining models?	(2.5)	
	(c) In a text, the first-ranked word appears 12,000 times. Estimate the total number of words used in the text assuming that the text follows Zipf's law up to rank 100.	(5)	
	UNIT-II		CO Mapping
Q4	(a) Define various Applications of Speech Processing in MDA.	(5)	(CO2)
	(b) In speech processing, the energy of a short speech signal segment is computed as the sum of the squared samples in that segment. Given the segment, S= [2,-1,3,-2]. Calculate the energy of the signal.	(5)	
Q5	(a) The spectral centroid represents the center of mass of a spectrum. Given the frequency bins, f= [100,200,300,400] and their corresponding magnitudes m = [1,3,5,2]. Calculate the Spectral centroid.	(5)	(CO2)

	(b) Define Formant frequencies, further solve the following A speaker has a vocal tract length of 17 cm. Assuming that the speed of sound in air is 34,000 cm/s, calculate the first three formant frequencies (F1, F2 , F3) for this speaker using the formula for an open-closed tube.	(5)	
UNIT-III			**CO Mapping**
Q6	(a) Explain the Neighborhood Processing Techniques in Multimodal Data Analysis. (b) Briefly Explain the challenges of Point Processing.	(5) (5)	(CO3)
Q7	(a) Given an audio signal with the following amplitude values at each time point: Amplitudes = [0.1, 0.3, 0.5, 0.2, 0.6, 0.9] Calculate the difference between each consecutive time point to detect sudden changes (edges) in amplitude. (b) Explain the benefits of Enhancement in MDA.	(5) (5)	(CO3)
UNIT-IV			**CO Mapping**
Q8	(a) What are the advantages and limitations of Conventional Multimodal Learning. (b) Explain the challenges in Processing Biomedical Signals?	(5) (5)	(CO4)
Q9	(a) Explain Spatial and Temporal Data Handelling. (b) What are the types, characteristics and applications of Biomedical Signals?	(5) (5)	(CO4)